Airframe Design Theory Exploration: Part 1

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Vocabulary

0.1 Wing Area

Denoted by S, wing area is the two dimensional area of the wingspan, as measured from above and projected onto a horizontal plane. It includes the combined area of both wings. One of the most common reasons to increase wing area is to decrease stall speed (see below for definition). The stall speed for level flight is given by:

$$V_s = \sqrt{\frac{2W}{C_{L_{max}} \rho S_{ref}}} \tag{1}$$

Increasing wing area leads to increased parasitic drag and wing weight. It also affects the optimal cruise lift coefficient, which may or may not be desirable.

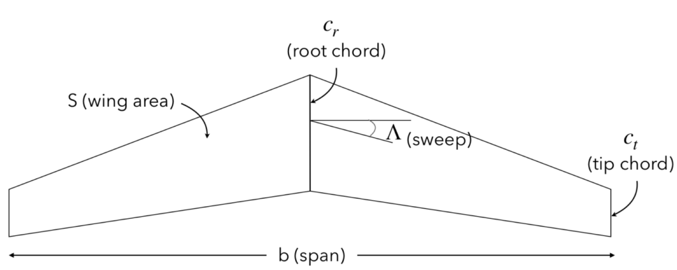


Figure 1: Depiction of Wing[1]

0.2 Chord Line

A straight line connecting the leading edge and the tailing edge of an airfoil[2]. This is important for finding the angle of attack, the lift coefficients, drag coefficients, etc, and can be changed mid flight through the use of flaps. The geometry of the airfoil does not affect the chord line given that the leading edge and tailing edge are located in the same positions. In other words, you can change the camber of an airfoil, and it won't affect the chord line as long as the leading and tailing edges remain in their original locations.

Note: Camber is the curvature of the airfoil. Positive camber means the mean camber line sits above the chord line. A symmetric airfoil would have zero camber.

[1] A. Ning, *Flight Vehicle Design*, 2018, pg. 31.
[2] A. Ning, *Flight vehicle Design*, 2018. pg. 27.

Mean Geometric Chord A representative chord line used for Reynolds number calculations and normalizing moments. Its found by finding the chord length of a hypothetical rectangular wing with the same wing area S and wingspan b as the original wing, and is denoted by $\bar{c}$.

$$\bar{c} = \frac{S}{b} \tag{2}$$

Mean Aerodynamic Chord The mean aerodynamic chord is a chord-weighted average chord, and is important for stability analysis. Portions of the wing with a larger chord have a larger impact on the mean aerodynamic chord than portions with a smaller chord.

$$c_{mac} = \frac{2}{3} \int_0^{b/2} c^2 dy \tag{3}$$

For a linearly tapered wing, the mean aerodynamic chord is:

$$c_{mac} = \frac{2}{3}(c_r + c_t - \frac{c_r c_t}{c_r + c_t}) \tag{4}$$

You can find the aerodynamic center of a wing by finding where its chord is equal to the mean aerodynamic chord.

0.3 Taper Ratio

The ration of the tip chord (See Figure 1) to the root chord, as defined by equation 5 (See figure 2b). This affects the lift distribution as well as the wing area. As the taper ratio decreases, the lift load distribution is weighted towards the root of the wing (more lift is being produced at the root than at the tip). This tends to reduce the amount of induced drag being produced. As the taper ratio increases, the lift load distribution is weighted towards the tip of the wing (more lift is being produced at the tip than at the root).

$$\lambda = \frac{c_t}{c_r} \tag{5}$$

0.4 Span

The span (b) , or wingspan, is the distance, from tip to tip of the airplane's wings. This distance does not factor out the width of the plane's fuselage (See figure 2). Span is always measured horizontally, and is not affected if the wing's dihedral angle is changed. Increasing a wing's span increases its aspect ratio (see below), which subsequently lowers the induced drag being produced. Decreasing the span increases induced drag but also increases the plane's maneuverability.

0.5 Aspect Ration

Aspect ration is defined by:

$$AR = \frac{b^2}{S} \tag{6}$$

where b is the length of the wingspan, and S is the wing area. b is the distance measured from wingtip to wingtip (See figure 2).

[3] A. Ning, *Flight Vehicle Design*, 2018, pg. 31.

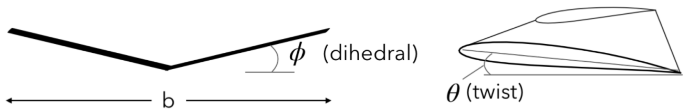


Figure 2: Depiction of Wing[3]

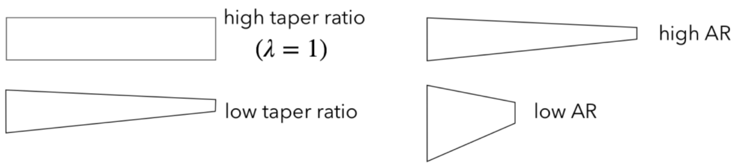


Figure 2b: Examples of a low and high taper ratio and a low and high aspect ratio.[4]

A high aspect ratio will result in long, skinny wings as shown in figure 2b. A low aspect ration will result in short stubby wings (often seen in fighter jets). Wings with a higher aspect ratio tend to have less induced drag, and have a higher lift-to-drag ratio. Wings with a lower aspect ratio tend to be more maneuverable but have more drag.

0.6 Sweep

The angle between the quarter chord line (or sometimes the leading edge) and a line perpendicular to the plane's fuselage. "The primary reason to use sweep is to decrease transonic wave drag"[5]. An aft-swept wing tends to improve high speed stability, while a forward swept wing tends to enhance maneuverability.

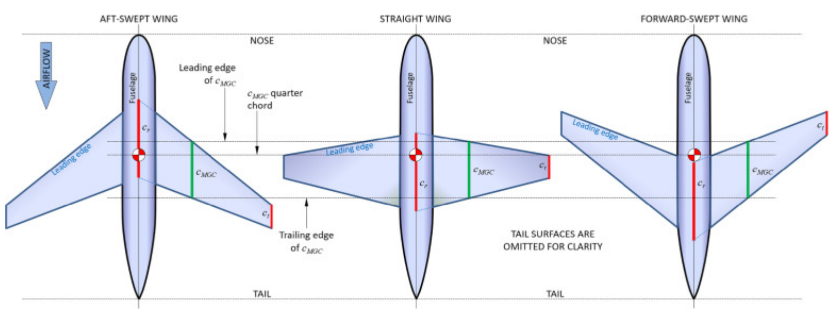


Figure 3: Types of Sweep for a Plane[6]

0.7 Dihedral Angle

The dihedral angle (Φ) is a measurement of how offset the wings are from horizontal. The angle is positive when the wings are offset above the horizontal line, and negative when below it. In figure 2, we see a positive Φ of approximately 15°. An increased Φ provides more stability for the aircraft, as the lift produces a restoring moment (the lift doesn't act only

[4] A. Ning, *Flight Vehicle Design*, 2018, pg. 32.
[5] A. Ning, *Flight vehicle Design*, 2018. pg. 76.
[6] Science Direct Topics, https://www.sciencedirect.com/topics/engineering/sweep-angle

along the y-axis, acting in part along the x-axis, pointing towards the fuselage); however, it decreases responsiveness.

0.8 Twist

Twist refers to the change in the wing's angle of incidence from root to tip. There are two main types of twist: root twist and tip twist. Tip twist is when the tip's angle of attack is different than the rest of the wing, and root twist is similarly when the root's angle of attack is different than the rest of the wing's angle of attack. These two types of twist are generally used for a linearly twisted wing. Washout (See below) can be produced when the tip has a lower angle of incidence than its root. The twist distribution throughout the wing can vary.

0.9 Washout

Washout describes when the tip of a wing has a lower angle of incidence than its root (while at zero lift). In other words, the tip of the wing is twisted downwards in relation to the root. This causes c_{max} to be positive, resulting in more stability for the aircraft by ensuring that the roots stall before the tips. This is important since the tips are further away from the plane's fuselage, therefore creating a larger moment when the forces acting upon them change. It can be difficult to design well; however, an appropriate combination of sweep, twist, taper, and aspect ratio can relatively efficiently achieve stability and trim[7]. Without washout, pilots would have less control over their plane and would be more likely to roll. It can also help reduce induced drag.

0.10 Lift

Lift is the force created by the pressure differential between the low-pressure (suction) surface and high-pressure (pressure) surface of the wing. This pressure differential is created by the flow of air around the shape of the airfoil, and this lifts the wing upwards. Lift is always defined as being perpendicular to the freestream direction. As the airstream passes by the plane, the airfoil separates the stream of air, creating a low-pressure zone above the wing and a low-pressure zone below it. The high pressure zone pushes the wing upwards, and as the airfoil's camber increases, this pressure difference grows, creating more lift. An increase in the airstream's flow speed will also generate an increased amount of lift; however, its important to note (as seen in figure 12) that the amount of lift produced depends greatly on the angle of attack α.

The lift coefficient of an airfoil is related to the area of the airfoil by the equation:

$$c_l = \int_0^1 (c_{p_l} - c_{p_u}) d\frac{x}{c} \tag{7}$$

where c_{p_l} and c_{p_u} are the pressure coefficient on the lower and upper surface respective, and $\frac{x}{c}$ is the normalized axial position along the airfoil from 0 to 1[8]. Optimized designs will be able to produce lift efficiently with minimal drag.

[7] A. Ning, *Flight vehicle Design*, 2018. pg. 93.
[8] A. Ning, *Flight vehicle Design*, 2018. pg. 40.

0.11 Drag

Drag is the aerodynamic force that opposes an aircraft's motion through the air. It acts opposite the direction of motion and is generated by the interaction between a solid body and the surrounding fluid (typically air). Its always defined as being parallel to the freestream direction (perpendicular to lift).

Because an airfoil is 2D, the drag is actually drag per unit span, represented by D'. While in 2D, the coefficient of drag is defined by:

$$c_d = \frac{D'}{q_\infty c} \tag{8}$$

where q_∞ is the dynamic pressure. A three-dimensional body, like a wing, is defined similarly but with an extra length dimension in the denominator:

$$c_d = \frac{D}{q_\infty S_{ref}} \tag{9}$$

The type of reference area S_{ref} used must be specified when providing coefficients. For a wing it is typically the planform area, whereas for a fuselage the cross-sectional area is typical. For a full aircraft the trapezoidal reference area is typically used; its not the exact same shape as the wing, but is easier to use. Any area can be used as long as it is consistent and is defined, it merely provides the length scales used in the normalization.[9].

Optimized airfoil designs will be able to produce lift efficiently while reducing drag. One form of drag is produced by shock waves (compressibility drag), and quickly becomes the primary source of drag in hypersonic aircraft; however, sharp leading edges will reduce the strength of these shock waves(see figure 4). For more information on these and other forms of drag, see below under 0.11.

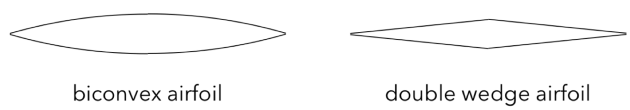


Figure 4: Two examples of airfoils used for locally supersonic flow.[10]

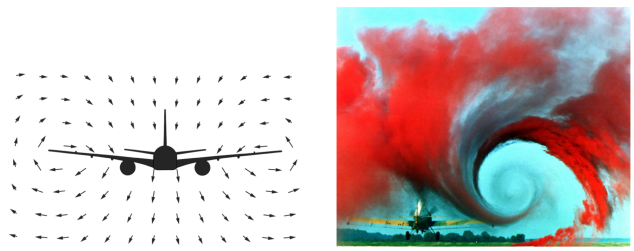
0.11.1 Induced Drag

Also known as vortex drag or lift-dependent drag, induced drag is the drag necessary in order to generate lift. Newton's third law requires that the upward force on the aircraft's wings be accompanied by an equal and opposite downward force exerted on the air underneath the wings. This means that the air underneath a wing is pushed downwards; however, instead of continuing downwards, this air circulates and creates a vortex around the wing (See figure 5a). This effect is not a tip effect, and will occur regardless of whether or not the wing has a tip, although it is true that circulation of the vortex is strongest at the wing's tip. Because of this, slow flight requires more power than fast flight. Winglets can modify the lift distribution

[9] A. Ning, *Flight vehicle Design*, 2018. pg. 36.
[10] A. Ning, *Flight Vehicle Design*, 2018, pg. 30.

in such a way that it acts like a span extension and increasing span decreases the amount of induced drag.

Induced drag is why you see horseshoe-shaped vortexes form behind an airplane (See figure 5b). These vortexes also limit the amount of planes that can use a runway at once (as you must wait for them to dissipate before takeoff).



(a) Visualization of airflow in plane of the wing.

(b) Picture of aircraft flying through colored smoke to visualize the wake vortex. Picture from NASA, public domain.

Figure 5: Visualization of wake vortex.[11]

0.11.2 Parasitic Drag

Two types of stresses act on any body in a fluid: normal stresses (pressure) and shear stresses (skin friction drag). Their contributions to drag are known as pressure drag and skin friction drag, and their sum is called parasitic drag, zero-lift drag, or sometimes just viscous drag. For many aircraft, skin friction drag is the largest component of drag; however it decreases as the plane's velocity increases (See figure 6).

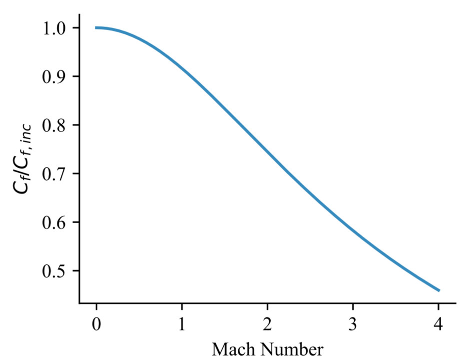


Figure 6: Reduction in skin friction coefficient at high Mach numbers.[12]

Skin Friction Drag Skin friction drag is the drag caused by viscous shear stresses acting over a body. For high Reynolds number flow, typical of most aircraft, the flow is essentially inviscid everywhere, except near the body (See figure 7). This region near the body where

[11] A. Ning, *Flight Vehicle Design*, 2018, pg. 56.
[12] A. Ning, *Flight Vehicle Design*, 2018, pg. 50.

viscous effects are important is called the boundary layer. Far away from the body the average speed of the molecules must be the freestream speed. But at the surface the average speed must be zero because the molecules are exchanging energy with the solid surface that is not moving (in the frame of reference of the solid object). Thus, a boundary layer forms to transition between the solid surface and the external velocity. Skin friction drag is caused by the interaction between the fluid and the body[13]. The skin friction drag, per unit length, is given by:

$$D' = \int_0^L \tau_w dx \tag{10}$$

If normalized (in the same manner as the drag coefficient), it is given the name skin friction coefficient and is represented by:

$$C_f = \frac{D'}{q_\infty L} = \int_0^1 \tau_w d(\frac{x}{L}) \tag{11}$$

where τ_w is the shear stress at the wall.

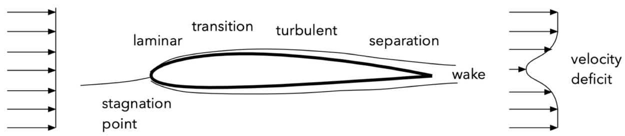


Figure 7: Depiction of a boundary layer in a high Reynolds number flow.[14]

For a laminar boundary layer, the skin friction coefficient is given by:

$$C_f = \frac{1.328}{\sqrt{Re}} \tag{12}$$

Turbulent boundary layers have no analytical solution; however, a simple version of an empirical fit is:

$$\frac{C_f}{C_{f_{inc}}} = (1 + 0.144M^2)^{-0.65} \tag{13}$$

It's important to note that an airfoil typically isn't a flat plate. We can approximate it for conceptual design; however, its typically best to use a combination of laminar and turbulent flow when calculating the skin friction drag.

Pressure Drag Pressure drag is the aerodynamic drag caused by a difference in pressure between the front and rear surfaces of an object moving through a fluid. An airfoil separates the stream of air, creating a low-pressure zone behind the wing. This difference in pressure causes the plane to be pushed backwards, albeit marginally. A flat plate at zero lift will have no pressure drag. As the airfoil's camber increases, this pressure difference grows, creating more drag. An increase in the airstream's flow speed will also generate an increased amount of pressure drag.

Note: Positive camber means the mean camber line sits above the chord line. A symmetric airfoil would have zero camber. Camber is the curvature of the airfoil.

[13] A. Ning, *Flight vehicle Design*, 2018. pg. 47.

[14] A. Ning, *Flight Vehicle Design*, 2018, pg. 30.

0.11.3 Compressibility Drag

At high speeds shock waves are formed, and energy is carried away by these shock waves. This type of drag is called compressibility drag or wave drag. For Mach numbers less than 0.3 compressibility drag is generally negligible. However, for commercial aircraft, which typically fly around Mach 0.75–0.9, or jets which can fly well above the speed of sound, this is an important if not dominant source of drag[15]. At speeds below mach 1, the air particles are given time to move around the airfoil, and will feel the pressure of its coming before it arrives; however, when traveling faster than mach one, they cannot and instead shock waves are generated, as shown in figure 8. These waves can be counteracted by sweeping the wings backwards; however, this can come with drawbacks (like a decrease in aircraft stability), which are generally undesired for commercial aircraft.

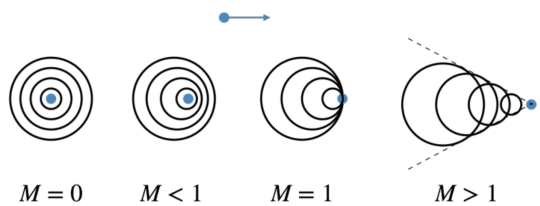


Figure 8: Sound waves emanating from a point moving to the right for four different Mach numbers.[17]

0.12 Pitching Moment

The pitching moment is the moment about the airfoil caused by the imbalance in how forces are applied throughout its surface. This is caused because of its non-symmetric geometry and the angle of attack (the angle between the freestream direction and the chord line. As forces are applied (drag and lift), they are not spread evenly throughout the airfoil due to its curvature and how the airstream flows around it. This generates a moment which must be balanced in order to maintain stability aboard the aircraft.

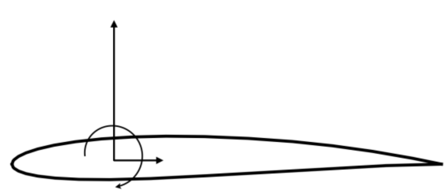


Figure 9: The pressures and shear over any body can be resulted into an equivalent set of forces and moments, two forces and one moment in 2D, and three forces and three moments in 3D.[19]

[15] A. Ning, *Flight vehicle Design*, 2018. pg. 46.
[17] A. Ning, *Flight Vehicle Design*, 2018, pg. 64.
[19] A. Ning, *Flight Vehicle Design*, 2018, pg. 33.

0.13 Lift and Drag Polars

Similar to the lift curve slope (see below), the drag polar is the variation in drag coefficient as compared with the angle of attack (or typically, the lift coefficient)[20]. Drag polars are graphs that plot how drag changes in relation to the lift coefficient. Lift polars are graphs that plot how lift changes in relation to the drag coefficient. As seen in figures 10 and 12, as you change the angle of attack, not only does it affect the amount of lift, but it also affects the amount of drag. Pilots can change the angle of attack of an airfoil mid-flight using flaps, giving the desired amounts of lift or drag at differing times (for example, while at cruising altitude versus takeoff). Figure 10 shows a notional drag polar; however, its typically plotted against the lift coefficient as opposed to the angle of attack.

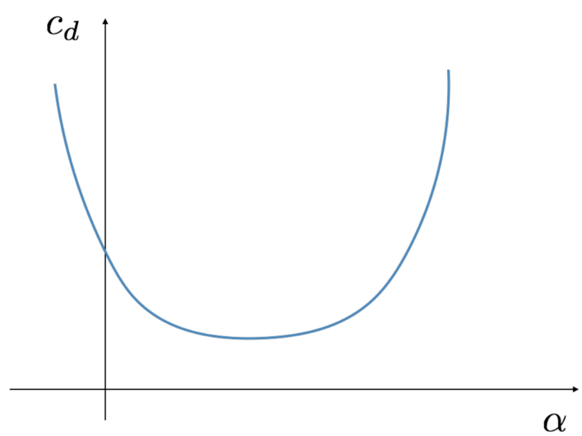


Figure 10: A notional drag polar. Plotted here versus angle of attack, but versus lift coefficient is more conventional.[22]

The Angle of Attack The angle of attack, denoted by α, is the angle between the freestream and the chord line. As shown in figure 12, the angle of attack significantly affects the lift coefficient, increasing it until reaching a maximum lift coefficient $c_{l_{max}}$ and then decreasing. Lift varies linearly with α until approaching stall. Twist is used in a wing to change the angle of attack in order to create washout at the desired locations (generally the roots). When the angle of attack is too large, the airstream no longer follows the curvature of the airfoil and will instead create a low-pressure zone behind the wing, generating large amounts of drag.

Mathematically, its equivalent to define the chord line as being horizontal (along the x-axis); however, in practice the chord line is typically rotated and the freestream is along the x-axis. This is simply done as a convenience when analyzing the airfoil's design.

Zero Lift Angle of Attack The zero lift angle of attack (α_0) is the angle of attack at which the airfoil will produce zero lift. A symmetric airfoil would have zero lift at an angle of attack of zero, thereby making it's zero lift angle of attack $\alpha = 0°$. Adding positive camber

[20] A. Ning, *Flight Vehicle Design*, 2018, pg. 42.
[22] A. Ning, *Flight Vehicle Design*, 2018, pg. 43.

causes the zero-lift angle of attack ($_0$) to become negative. That means that at an angle of attack $\alpha = 0°$, a positively cambered airfoil produces lift. The flaps on an airplane's wing allow for manual control over the angle of attack, allowing the pilot to more greatly control how much lift is being produced.

Note: Positive camber means the mean camber line sits above the chord line. A symmetric airfoil would have zero camber. Camber is the curvature of the airfoil.

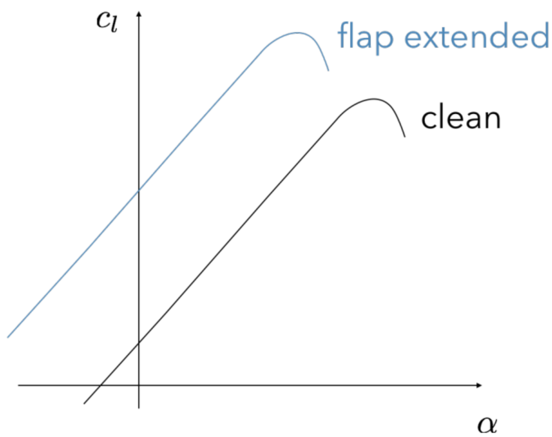


Figure 11: Flaps change the effective camber, and thus the zero lift angle of attack. They have almost no effect on $c_{l_{max}}$ however.[24]

Using flaps to give the wing a positive camber increases the amount of lift produced while the plane's body remains at the same angle of attack relative to the airstream. The maximum angle of attack (the angle of attack that will produce the most lift) will subsequently decrease (See figure 11). This is helpful during takeoff.

Lift Curve Slope The force/moment curves for a given airfoil are a function of the angle of attack (α), Reynolds numbers, and Mach number. When operating within a narrow range of Re and M in which varying these parameters does not significantly change the airfoil's performance, the lift, drag, and moment coefficients effectively become only a function of the angle of attack. In this scenario, the lift coefficient can be plotted against the angle of attack, producing a lift curve, as shown in figure 12. The lift curve slope is specifically the slope of the linear portion of the lift curve, and is usually denoted by a m or an a. As the effect α has on the lift coefficient increases, the steeper the lift slope curve will be.

Stall Stall occurs because the angle of attack becomes too large and the flow separates from the airfoil, causing a significant increase in drag and a decrease in lift[27]. When the angle of attack is too large, the airstream no longer follows the curvature of the airfoil and will instead create a low-pressure zone behind the wing, generating large amounts of drag. The maximum lift coefficient that can be achieved before stall is denoted by $c_{l_{max}}$ or $C_{L_{max}}$ in 3D. In figure 12, you can see this by observing where $c_{l_{max}}$ is achieved, and the coefficient of lift begins to

[24] A. Ning, *Flight Vehicle Design*, 2018, pg. 143.
[27] A. Ning, *Flight vehicle Design*, 2018. pg. 41-42.

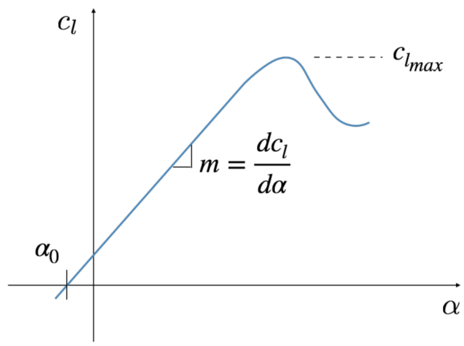


Figure 12: A notional lift curve slope highlighting the zero-lift angle of attack, lift curve slope, and maximum lift coefficient before stall.[26]

decrease. For a 2D airfoil, the lift coefficient prior to stall can be computed as:

$$c_l = m(\alpha - \alpha_0) \tag{14}$$

where m is the lift curve slope. Critical wing theory is also used to determine when stall might occur. A wing should optimally stall at the roots before it stalls at the tips, as this increases stability, and you can control this by changing the wing's twist to create washout. Optimally, a wing should stall at the roots first, as the tips are further away from the plane's fuselage, therefore creating a greater moment about the plane when they do stall, decreasing stability.

[27] A. Ning, *Flight Vehicle Design*, 2018, pg. 42.

Exploration

0.14 Code

```
using VortexLattice
using Plots
using LinearAlgebra

#THIS SECTION DESCRIBES THE WING GEOMETRY
#These are the coordinates for two separate locations that are joined by a straight line, combined into arrays
xle = [0.0, 0.4] # leading edge x-position.
yle = [0.0, 7.5] # leading edge y-position
zle = [0.0, 0.0] # leading edge z-position
#The wing's leading edge goes from (0,0,0) to (0.4,7.5,0). It's important to note that this is only
#for a single wing, and the two data points for chord length (below) are for the root and the tip chord lengths, respectively

chord = [2.2, 1.8] # chord length
theta = [2.0*pi/180, 2.0*pi/180] # twist (in radians), for the root and tip, respectively.
phi = [0.0, 0.0] # section rotation about the x-axis. In other words, its the dihedral angle (in radians)
#(The measurement of how offset the wings are from the horizontal plane of the aircraft.)

fc = fill((xc) -> 0, 2) # camberline function for each section (y/c = f(x/c))
#This function describes the camberline of the airfoil sections along the span of the wing, creating the upper and
#lower edges of the airfoils. This function is set to zero (the first number), indicating that the wing has no camber
#and is a flat plate. The two indicates that there are two sections (root and tip) for which this function is defined
#(if xle, yle, & zle had three points, you would increase this second digit to be three).

ns = 12 # number of spanwise panels. Each panel is a trapezoidal shape bounded by the chords at the inner and outer
#edges, as well as the leading and trailing edges of the wing. Each panel has a different shape.
nc = 6  # number of chordwise panels. Similar to spanwise panels, but in the chordwise direction.
spacing_s = Sine() # spanwise discretization scheme. This scheme uses a sine function to distribute the panels along
#the span of the wing, resulting in a higher concentration of panels near the wing root and tip, where aerodynamic
#effects are the greatest. In other words, the panels closest to the root and tips are narrower than those in between
#them (in the spanwise direction).
spacing_c = Uniform() # chordwise discretization scheme. Insted of concentrating the panels near the edges (See
#Sine function above), the panels are uniform throughout the wing (in the direction of the chord line).

grid, ratio = wing_to_grid(xle, yle, zle, chord, theta, phi, ns, nc;
fc = fc, spacing_s=spacing_s, spacing_c=spacing_c, mirror=true)#This combines all the factors included in the ()
#in order to create a meshed, 3-D shape of the wing. It stores this shape in the variable "grid", which is used later
#in the code to perform aerodynamic analysis. The "ratio" variable is a vector storing chorwise panel stretching ratios
#(the shape of the panels in the chordwise direction). This is used to compute the aerodynamic influence matrices
#and the scale vortex strengths during the analysis. The mirror option is set to true, meaning that the wing is mirrored
#across the aircraft's centerline to create a full wing (instead of just a half-wing).

grids = [grid] #list of grids in the system (only one wing in this case). It simply stores the grid created above
#in a vector(useful for organizing and managing multiple grids in a more complex analysis involving several lifting
#surfaces)
ratios = [ratio] #list of ratios in the system (only one wing in this case). It simply stores the grid created above
#in a vector(useful for organizing and managing multiple grids in a more complex analysis involving several lifting
#surfaces)
system = System(grids; ratios) # Combines grids and ratios in a System object, which is a data structure used to store
#and manage all the lifting surfaces in the analysis. This object is essential for performing aerodynamic calculations.

Sref = 30.0 # reference area
cref = 2.0  # reference chord
bref = 15.0 # reference span
rref = [0.50, 0.0, 0.0] # reference location for rotations/moments (typically the center of gravity)
Vinf = 1.0 # reference velocity (magnitude)
ref = Reference(Sref, cref, bref, rref, Vinf) #Creates a Reference object that stores the reference parameters used
#in the aerodynamic analysis.

alpha = 1.0*pi/180 # angle of attack
beta = 0.0 # sideslip angle. The angle between the aircrafts forward pointing axis and the actual direction the air
#is coming from. When you set beta = 0.0, you are telling the solver the aircraft has no sideways motion relative to
#the airflow.
Omega = [0.0, 0.0, 0.0] # rotational velocity around the reference location (rref). This vector represents the angular
#velocity of the aircraft in three-dimensional space, with components along the x, y, and z axes. Setting it to
#[0.0, 0.0, 0.0] indicates that the aircraft is not rotating.
fs = Freestream(Vinf, alpha, beta, Omega) #Combines the freestream parameters into a Freestream object, which is used
#to define the incoming airflow conditions for the aerodynamic analysis.

symmetric = false #When set to false, the solver considers both sides of the aircraft separately, allowing for asymmetric
#flow conditions and forces. This is important for accurately capturing the aerodynamic behavior of the aircraft,
#especially during maneuvers (anytime theres a rolling and yawing moment) or in the presence of crosswinds. When set
#to false, the solver does not assume symmetry about the aircraft's centerline and cancel out forces and moments.
```

```
steady_analysis!(system, ref, fs; symmetric)
#This function performs a steady-state aerodynamic analysis on the lifting surfaces defined in the "system" object,
#using the reference parameters in the "ref" object and the freestream conditions in the "fs" object (Simply put,
#it takes all the information found above, and performed an aerodynamic analysis). The analysis assumes that the flow
#conditions are constant over time (steady) and that the aircraft is not symmetric about its centerline (symmetric=false)
#The results of the analysis, including aerodynamic forces and moments, are stored within the "system" object for further
#evaluation
CF, CM = body_forces(system; frame=Wind()) #This creates the force and moment vectors in the wind frame of reference.
#Of the force vector, the x-component is the drag force, the y-component is the side force, and the z-component is
#the lift force. Of the moment vector, the x-component is the rolling moment, the y-component is the pitching moment,
#and the z-component is the yawing moment.

# extract aerodynamic forces from the vectors created above.
CD, CY, CL = CF
Cl, Cm, Cn = CM

CDiff = far_field_drag(system)#Computes the drag contribution from the far-field induced drag, which is the drag caused b
#vortices shed into the wake behind the lifting surfaces. This function calculates the far-field drag based on the vortex
#and wake geometry determined during the steady-state analysis. The result is stored in the variable "CDiff".
#Simply put --> It calculated the coefficient of induced drag.

# calculate lifting line coefficients
cf, cm = lifting_line_coefficients(system; frame=Body()) #cf is the calculated section force coefficients along the
#span, and cm is the calculated section moment coefficients along the span. They are distributions and not total forces.
#frame=body means that its using the reference frame of the aircraft body (as opposed to wind frame used earlier).

dCFb, dCMb = body_derivatives(system)#This function computes the body-axis aerodynamic derivatives for the lifting
#surfaces defined in the "system" object. These derivatives quantify how the aerodynamic forces and moments change
#in response to small perturbations in the aircraft's motion, such as changes in velocity or angular rates. The
#results are stored in the variables "dCFb" and "dCMb", which contain the derivatives of the force and moment
#coefficients, respectively.
#SIMPLIFIED--> It calculated how the forces and moments change as the aircraft moves in different ways.

# traditional names for each body derivative. The following is simply separating the results found above into
#more manageable variables.
CXu, CYu, CZu = dCFb.u
CXv, CYv, CZv = dCFb.v
CXw, CYw, CZw = dCFb.w
CXp, CYp, CZp = dCFb.p
CXq, CYq, CZq = dCFb.q
CXr, CYr, CZr = dCFb.r
Clu, Cmu, Cnu = dCMb.u
Clv, Cmv, Cnv = dCMb.v
Clw, Cmw, Cnw = dCMb.w
Clp, Cmp, Cnp = dCMb.p
Clq, Cmq, Cnq = dCMb.q
Clr, Cmr, Cnr = dCMb.r

dCFs, dCMs = stability_derivatives(system) #This is doing the same thing as the body derivatives function above, but
#instead using the stability axes. The stability axes are a coordinate system aligned with the aircraft's
#longitudinal and lateral directions, which are more relevant for stability and control analysis. (This axis changes
#with the angle of attack and sideslip angle of the aircraft, making it more suitable for evaluating stability as it
#aligns more naturally with lift and drag forces.)

# traditional names for each stability derivative. The following is simply separating the results found above into
#more manageable variables.
CDa, CYa, CLa = dCFs.alpha
Cla, Cma, Cna = dCMs.alpha
CDb, CYb, CLb = dCFs.beta
Clb, Cmb, Cnb = dCMs.beta
CDp, CYp, CLp = dCFs.p
Clp, Cmp, Cnp = dCMs.p
CDq, CYq, CLq = dCFs.q
Clq, Cmq, Cnq = dCMs.q
CDr, CYr, CLr = dCFs.r
Clr, Cmr, Cnr = dCMs.r

write_vtk("simplewing", system) #This simply exports the results and the resulting 3-D model into a VTK file that you
#can open with visualization software like ParaView or VisIt. This allows you to visually inspect the geometry,
#paneling, and the aerodynamic results on the lifting surfaces.



##Lift vs Angle of Attack Curve

lift = []
angle = []
```

```
    global alpha = 1.0*pi/180 # angle of attack
for i in 1:500
    fs = Freestream(Vinf, alpha, beta, Omega)
    steady_analysis!(system, ref, fs; symmetric)
    CF, CM = body_forces(system; frame=Wind())
    push!(lift, CF[3])
    push!(angle, alpha * (180/pi))
    global alpha += 0.2*pi/180
end
using Plots
plot(angle, lift, xlabel="Angle of Attack (degrees)", ylabel="Lift",
    title="Lift Curve", legend=false)




# ###Induced drag vs Angle of Attack Curve
# induced_drag = []
# angle2 = []
#     global alpha = 1.0*pi/180 # angle of attack
# for i in 1:500
#     fs = Freestream(Vinf, alpha, beta, Omega)
#     steady_analysis!(system, ref, fs; symmetric)
#     CDiff = far_field_drag(system)
#     push!(induced_drag, CDiff)
#     push!(angle2, alpha * (180/pi))
#     global alpha += 0.2*pi/180
# end
# plot(angle2, induced_drag, xlabel="Angle of Attack (degrees)", ylabel="Induced Drag Coefficient",
#     title="Induced Drag Curve", legend=false)




# ##Moment vs Angle of Attack Curve
# momentroll = []
# momentpitch = []
# momentyaw = []
# moment = []
# angle3 = []
#     global alpha = 1.0*pi/180 # angle of attack
# for i in 1:1000
#     fs = Freestream(Vinf, alpha, beta, Omega)
#     steady_analysis!(system, ref, fs; symmetric)
#     CF, CM = body_forces(system; frame=Wind())
#     push!(momentroll, CM[1])
#     push!(momentpitch, CM[2])
#     push!(momentyaw, CM[3])
#     push!(moment, norm(CM))
#     push!(angle3, alpha * (180/pi))
#     global alpha += 0.1*pi/180
# end
# plot(angle3, momentpitch, xlabel="Angle of Attack (degrees)",
#     title="Moment Curves", legend=false, label="Pitching Moment", color=:blue)
# plot!(angle3, momentroll, legend=true, color=:brown, label="Rolling Moment")
# plot!(angle3, momentyaw, legend=true, color=:green, label="Yawing Moment")
# plot!(angle3, moment, legend=true, color=:purple, label="Total Moment")




# ###Lift vs Drag Curve
# lift2 = []
# drag2 = []
# for i in 1:250
#     global alpha = i * (0.2 * pi/180) # angle of attack
#     fs = Freestream(Vinf, alpha, beta, Omega)
#     steady_analysis!(system, ref, fs; symmetric)
#     CF, CM = body_forces(system; frame=Wind())
#     CDiff = far_field_drag(system)
#     push!(lift2, CF[3])
#     push!(drag2, CF[1])
# end
# plot(drag2, lift2, xlabel="Drag", ylabel="Lift",
#     title="Lift vs Drag Curve", legend=false)


# ##Lift/Drag vs Angle of Attack Curve
```

```
# lift = []
# angle5 = []
# drag3 = []
#     global alpha = 1.0*pi/180 # angle of attack
# for i in 1:500
#     fs = Freestream(Vinf, alpha, beta, Omega)
#     steady_analysis!(system, ref, fs; symmetric)
#     CF, CM = body_forces(system; frame=Wind())
#     push!(lift, CF[3])
#     push!(angle5, alpha * (180/pi))
#     push!(drag3, CF[1])
#     global alpha += 0.2*pi/180
# end
# plot(angle, lift, xlabel="Angle of Attack (degrees)", ylabel="Lift",
#     title="Lift Curve", legend=true, label="Lift")
# plot!(angle5, drag3, color=:red, label="Drag")
```

0.15 Graphs

Below are the graphs produced with this code. The values of all input are form the example problem in the GettingStarted portion of VortexLattice's instruction manual.

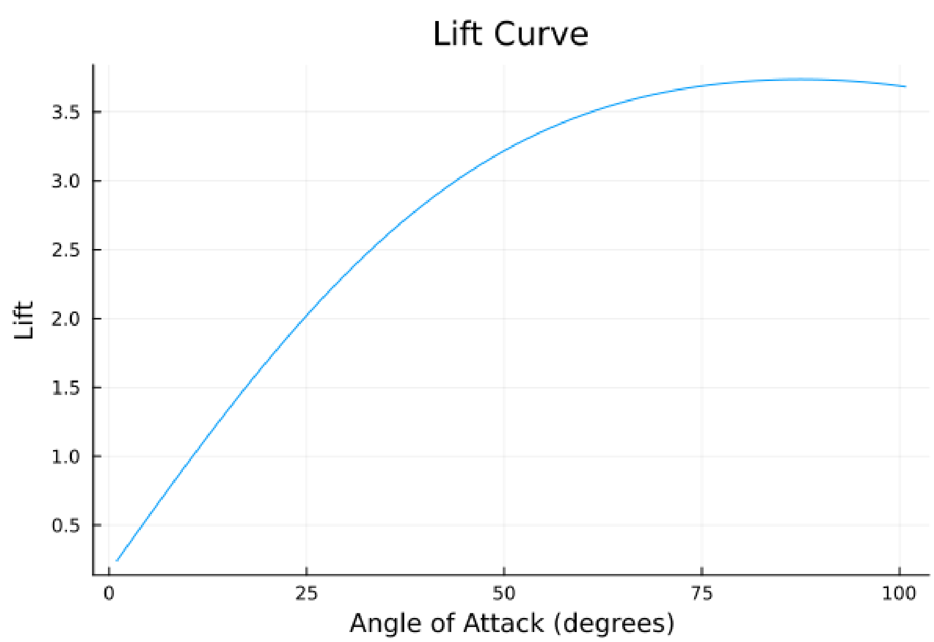


Figure 13: A description of how lift varies with regard to the angle of attack.

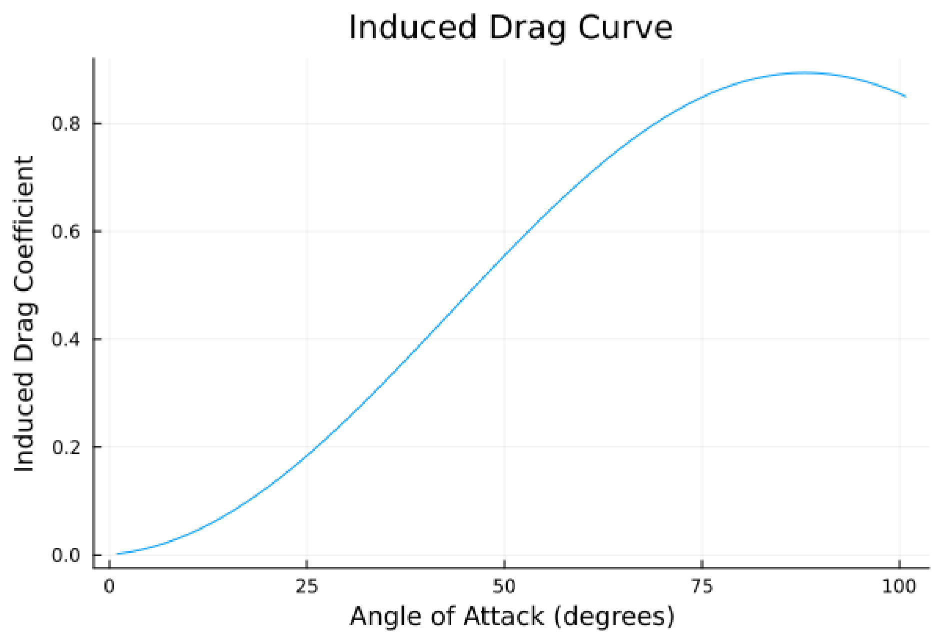


Figure 14: A description of how induced drag varies in regards to the angle of attack.

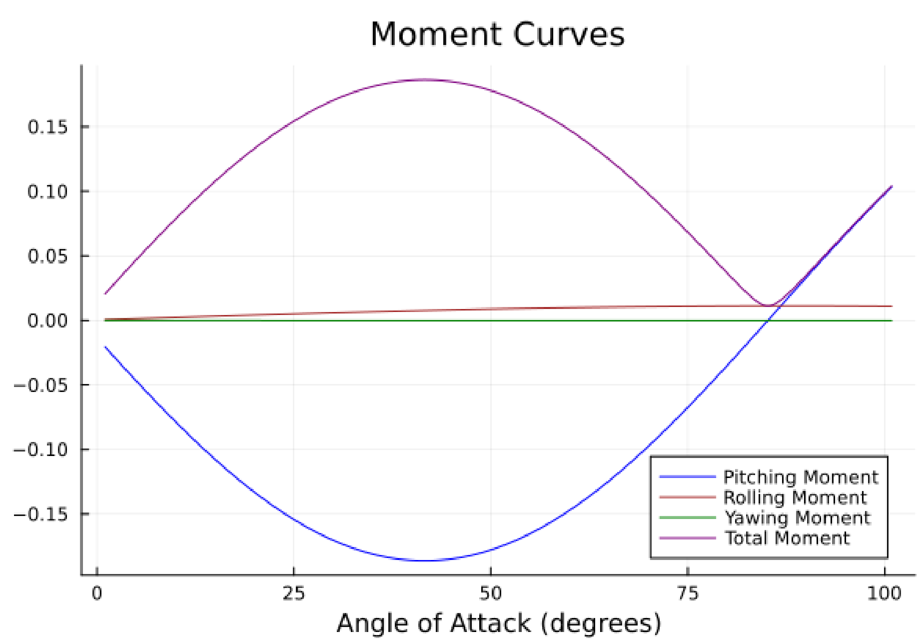


Figure 15: Different moments about the rref location ([0.50, 0.0, 0.0]).

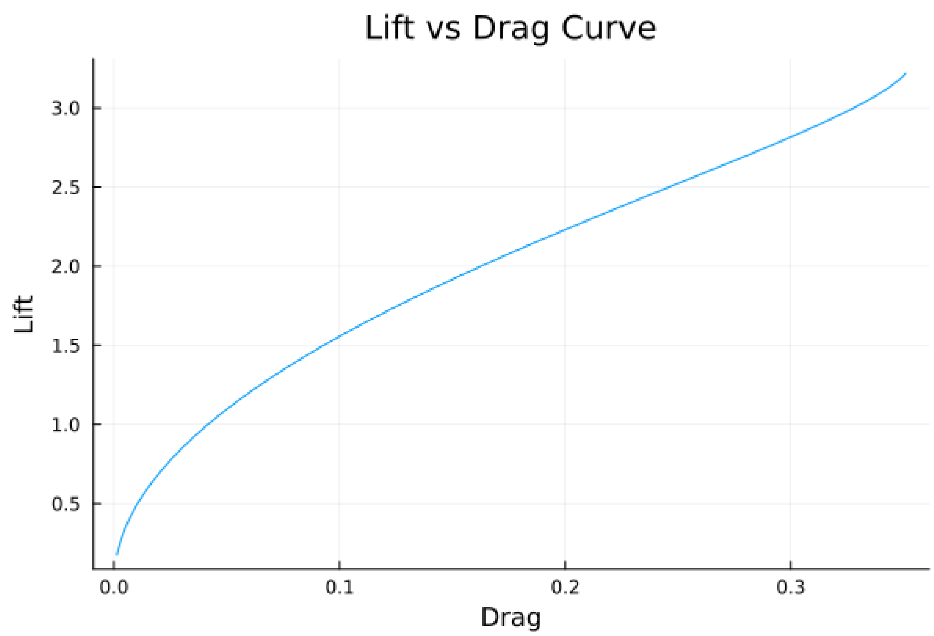


Figure 16: A description of how lift varies with regards to drag.

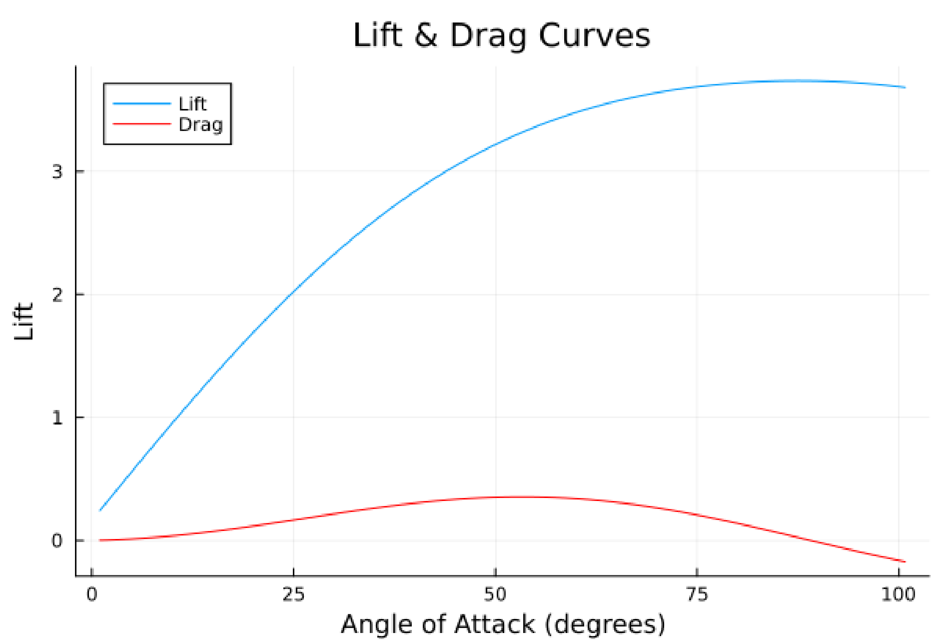


Figure 17: A description of how lift and drag vary in regards to the angle of attack.

0.16 Discussion

Variation from the Thin Airfoil Theory According to the Thin airfoil theory, the lift curve slope of a thin 2D airfoil is theoretically 2π, or 6.28. In the example above (See Figure 13), the lift curve slope ended up being approximately 4.58, which is off by a 28 percent margin of error. This is significantly large, and could result in inaccurate models. There are several reasons why these numbers vary so greatly, and the thin airfoil theory doesn't account for 3D effects, working only to predict 2D airflow. In contrast, VortexLattice predicts some 3D airflow effects, which greatly increase the accuracy of the lift curve slope compared to how planes fly in reality. The thin airfoil theory simply can't account for viscous effects.

Its also important to note that for the thin airfoil theory, it assumes an infinite span; however, the span of our wings in this example is only 15 meters. If you were to increase the span, it would more closely resemble the thin airfoil theory. If you were to increase the Aspect Ration (AR) of the example above, it would also reduce the impact 3D effects have on the lift curve slope, therefore aligning it more with the thin airfoil theory's prediction.

Example Data Variation from Real-World Data In Figure 13, you'll notice that the amount of lift produced starts to decrease with an angle of attack of about 90° (for a symmetric airfoil with zero camber). This would imply that you get the most lift when your wings are almost completely vertical. As we know, this isn't necessarily the case. The data in the example above varies from real-world data due to several reasons. Its important to note that VortexLattice assumes the airflow is inviscid; however, in reality, air flows are viscous and will have boundary layers, causing separation at higher angles of attack. The larger the angle of attack, the greater these effects will be, and therefore the more skewed these graphs are. In order to minimize the error, maintain a lower angle of attack if using VortexLattice (normally it must be under twenty degrees). In reality, stall tends to occur above twenty degrees.

The results from Figure 13 only begin to decrease around 90°; however, a typical lift-curve graph would look more like this:

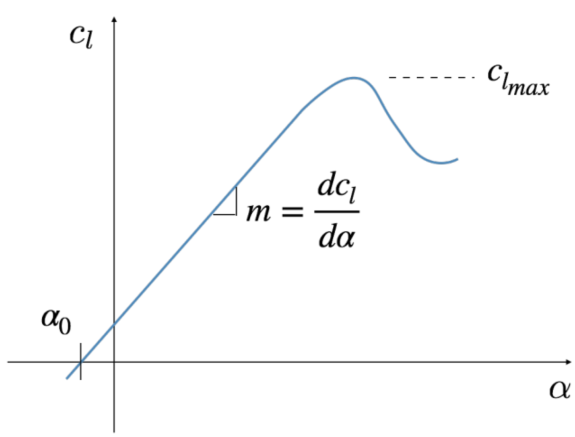


Figure 18: Flaps change the effective camber, and thus the zero lift angle of attack. They have almost no effect on $c_{l_{max}}$ however.[29]

VortexLattice also assumes that the airstream's air is incompressible, which is only the case for water, not air. It also can't simulate skin friction drag, pressure drag, form drag, and interference drag.

Trends in Example Plots There are several trends between the plots generated in the example problem above. Both induced drag and the lift curve begin to decrease with an angle of attack of approximately 90°; however, as discussed above, this would typically occur around 20° (See above for more information on this topic). Drag, on the other hand, begins to decrease around $\alpha = 45°$, which is similar to the pitching moment and total moment curves. This relationship is due to the fact that the moment curves are directly dependent on the forces acting on the wings (such as drag). In Figure 16, we see that lift and drag are positively correlated, and that lift increases as drag increases. This is because lift and drag are both dependent upon the angle of attack, and both positively increase as α increases (noting once again that this simulation isn't completely accurate due to stall and viscous airflow). The trends in these graphs are generally accurate when compared to real-world data; however, their exact values aren't fully accurate.

[29] A. Ning, *Flight Vehicle Design*, 2018, pg. 42.